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Characterization of MedConcept Limits to a 2D BiomedCLIP Adaptation.

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Medical VLMs are accurate but opaque.

The internal representations are high-dimensional, polysemic and unreadable. A serious concern in safety-critical clinical use.

§1 Introduction

Concept-based XAI: from supervised to unsupervised discovery.

Supervised

Concept Bottleneck Models (Koh 2020) and TCAV (Kim 2018) need a pre-defined, labelled concept set — exactly the limitation unsupervised discovery targets.

MedConcept (Haque 2026)

SAE extraction → alignment to a radiology vocabulary → an LLM-judge with Aligned/Unaligned/Uncertain metrics.

Impl: `src/autoencoder/`, `src/vocabulary_building/`, `src/evaluate_llm_judge.py`

Sparse Autoencoders

Bricken et al. (2023): dictionary-learning on activations yields monosemantic latents. The Top-K SAE (Gao 2024) sets L_0 exactly via top-k gating. We adopt it.

Six gaps this work addresses.

GAP 1

Instability / non-identifiability

Sparse factorisations are not reproducible across seeds; no prior medical-VLM work reports seed-stability against a null.

GAP 3

Representation-location mismatch

SAE literature uses raw hidden states; reference pipelines fit on the already-projected, gap-bearing CLIP space.

GAP 5

Flat concepts

Explanations are top-k lists that ignore anatomical / hierarchical structure.

GAP 2

Clinical validity

A cosine-assigned name is cosmetic unless it tracks real pathology.

GAP 4

Small-data robustness

SAEs need 10^5 – 10^6 activations; licence-free medical corpora are 2–3 orders smaller.

GAP 6

Non-reproducible evaluation

Claims rest on single metrics whose chance floor is never reported; slot-wise overlap is degenerate under per-seed permutation.

Our Adapted Architecture.

Instead of 3D Merlin, we use the 2D BiomedCLIP as backbone. Its image encoder exposes a **768-d hidden state** that a frozen projection compresses into the **512-d** contrastive space. Features are named against **domain-matched** vocabularies: **RadLex** for IU X-Ray (chest-specific), **MeSH** for ROCov2 (multimodal). Before of this, we correct the **modality gap**.

Impl: `src/embedding_extraction/extract_embeddings.py`, `src/vocabulary_building/build_vocabulary.py`

Two scales: IU X-Ray and ROCov2 (10×).

IU X-RAY · 1× · PRIMARY

The small-scale setting.

Split **by radiographic study**, so no study appears in both train and test — verified **zero** overlap, recomputed deterministically every run.

ROCOV2 · 10× · REPLICATION

A ten-times larger corpus.

Split by image. It exists to test one hypothesis head-on: **does more data cure non-identifiability?**

Impl: `src/utils.py` (split_embeddings) · anti-leakage test: `tests/unit/test_split.py`

Three decompositions, one SAE config.

01 · BASELINE

Top-K SAE · 512-d

Faithful MedConcept re-impl on the projected space. Multi-seed. (Gap 1)

02 · PATH A

Top-K SAE · 768-d

CLS hidden state *before* projection; bridge with frozen W_proj. (Gap 3)

03 · PATH B

SPLiCE

Fixed domain-matched dictionary (RadLex / MeSH) — no learning, deterministic. (Gap 1+4)

ONE SAE CONFIG FOR CLEAN COMPARISON

DICTIONARY D	TOP-K	STEPS	SEEDS
2048	32	8,000	5 (0, 42, 123, 456, 789)

Impl: `src/autoencoder/sae_module.py` (SAEManager), `src/sae_hidden/`, `src/concept_discovery/spliece.py`

A permutation-invariant matched metric.

For each decoder row of seed A, take its **best cosine match** across *all* rows of seed B, then average. Feature order does not matter — that is what makes it a real stability measure.



Best-match cosine, averaged over rows. Permutation-invariant by construction.

Impl: `src/autoencoder/stability_analysis.py`

A null conditioned on effective rank.

Real decoder directions live in a **subspace** of effective rank ~ 357 — not the ambient 512. We draw null vectors *inside* that subspace, so the comparison is honest. An isotropic full-512 null would inflate the ratio.

SUBSPACE-CONDITIONED NULL

	IU X-RAY	ROCOV2
Ambient dimensionality	512	512
Decoder effective rank	~ 357	~ 363
Null drawn in	top-rank subspace	top-rank subspace

Impl: `src/sae_hidden/naming_hidden.py` (`compute_random_baseline`)

5 · EXPERIMENTS AND ANALYSIS · THE HEADLINE RESULT · GAP 1

1.81×

1.67× IUX-Ray · 1.81× ROCov2

The baseline SAE reconstructs its inputs well — but cross-seed **feature identity is absent**: best-match barely above the conditioned null. Genuine non-identifiability, not a training bug.

Good reconstruction, absent identity.

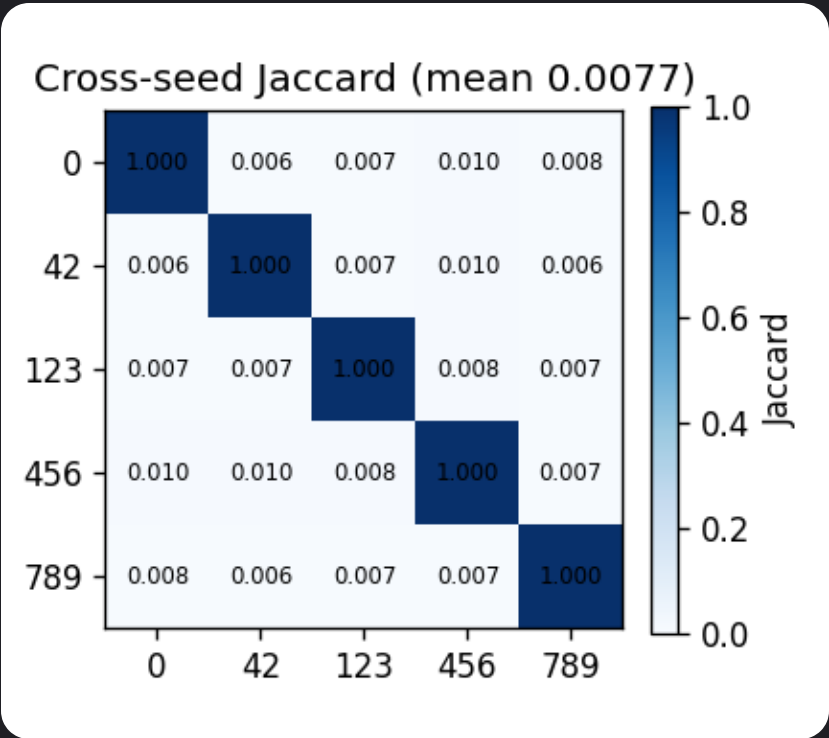
CROSS-SEED STABILITY, BASELINE TOP-K SAE

INDICATOR	IU X-RAY	ROCOV2 (10×)
Reconstruction cosine	0.99	0.97
Matched obs / null	1.67×	1.81×
Matched cosine (raw)	0.299	0.327
Features matching ≥0.9	0.0%	0.3%
Naming (top-1 cos)	0.40	0.48

Slot-wise is floor-by-construction.

Rename one SAE's features with a random permutation — **same network**, same reconstruction — and slot-wise Jaccard collapses to the **0.0077** floor. The matched metric correctly sees identity.

Impl: `scripts/relabeling_control.py`

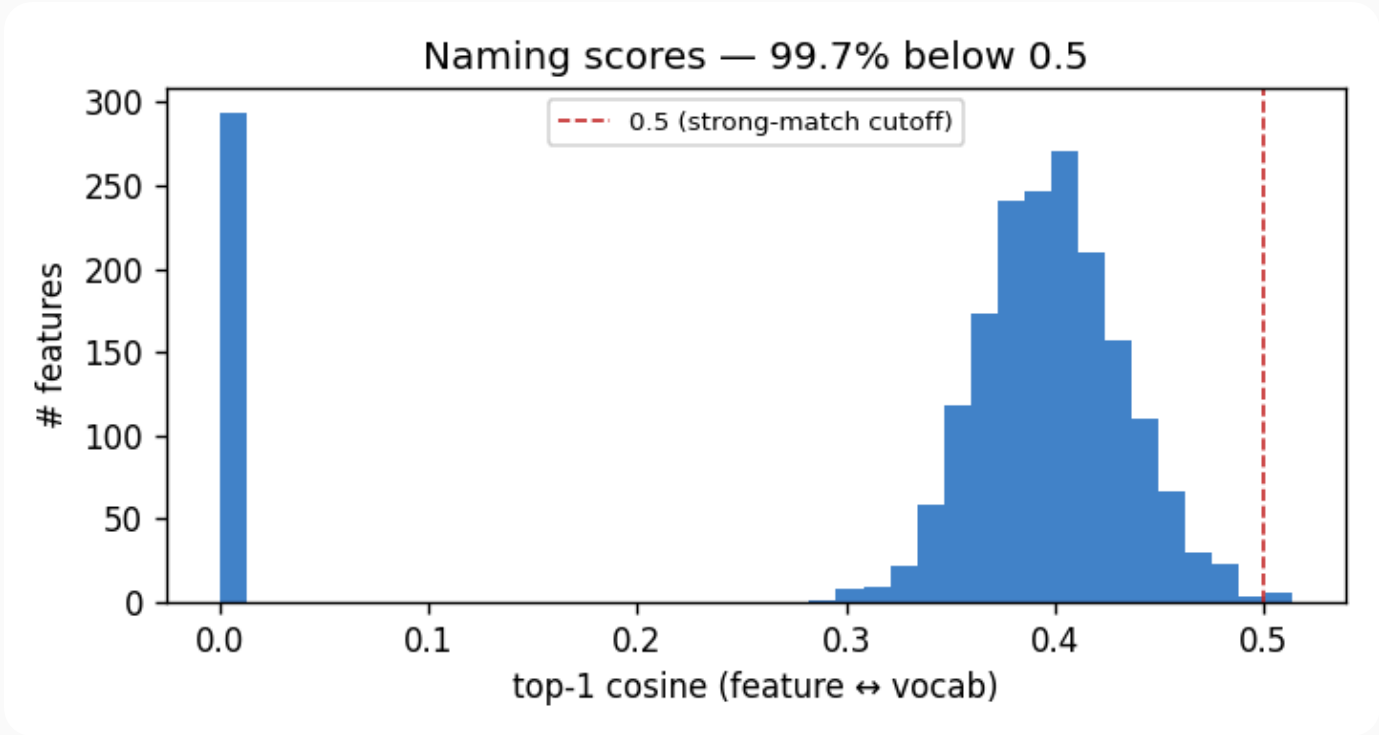


Cross-seed Jaccard (IU baseline). Mean ≈ 0.0077 — the analytical floor.

A healthier decomposition — same verdict.

Path A trains on the hidden state **before** the projection. Fewer dead features, better naming — but the matched verdict is unchanged: **weak universality**, a shared subspace, not identical features. Still the pooled CLS token.

Impl: `src/sae_hidden/`, `scripts/run_path_a.py`



Naming scores — top-1 feature↔vocab cosine. ~99.7% below 0.5 (max 0.513).

No overcompleteness escapes weak universality.

MATCHED OBS / NULL VS DICTIONARY OVERCOMPLETENESS

DICTIONARY D	TOP-K	OBS / NULL
1024	16	2.78×
2048	32	2.63×
4096	64	2.00×

More overcomplete → ratio drops toward the null. No setting is identifiable.

5 · EXPERIMENTS AND ANALYSIS · THE SCALE HYPOTHESIS · GAP 4

10_x

ROCOv2 ~80k vs IU X-Ray ~7k

Ten times more data improves **training health** (dead 16% → 0.6%) but **not identifiability** — matched 0.299 → 0.327, ratio 1.67× → 1.81×. Binding constraint = **representation site**, not volume.

Deterministic. Judge-ready.

Decompose each embedding into a sparse, non-negative combination over a **fixed domain-matched dictionary** (RadLex / MeSH). Never estimated → **deterministic by construction**, zero cross-seed instability.

- Fixed dictionary — no estimation, no seeds, no instability.
- Same output schema as the SAE — the same judge scores it.
- Honest caveat — frequent terms are "mixed" (a dictionary property).

Impl: `src/concept_discovery/spliece.py` – `decompose_image()`

Scales to both corpora. Gap is the fix.

SPLICE COVERAGE AND MODALITY-GAP CORRECTION

	IU X-RAY	ROCOV2
Dictionary	RadLex · 1031	MeSH · 1024
Terms used	997	1024
Images decomposed	1,515	15,958
Runtime	9.5 s	101 s
Max atom  img cos (gap-corrected)	0.49 → 0.85	0.54 → 0.86

Dense, but weakly separated.

Cluster the active vocabulary into **concept families** via ontology ancestors (RadLex / MeSH) + two degeneracy guards: leaf-root rejection and a subtree canopy that rejects overly generic ancestors. SPLiCE cuts concepts-per-image **~1.75×**.

ORGANISATION RESULTS · TAB.4

METHOD	CONCEPTS	FAMILIES	SILHOUETTE
SPLiCE (IU)	997	32	0.020
SAE baseline (IU)	80	9	0.094
Path A hidden (IU)	14	4	0.284
SPLiCE (ROCOv2)	1024	32	0.066

Impl: `src/concept_discovery/organize.py` · Path A hidden: 1260/1515 images empty

Three judges, three different biases.

MedGemma 4B

Domain-specialised medical LLM. Exhibits a **systematic positive bias**, rating all methods (including the random null control) at ~80% Aligned.

Gemma 4 26B

Large general-purpose LLM (4-bit). Shows a **strong rejection bias**, rejecting 87% of all baseline pairs as Unaligned.

Sellergren et al. 2025 — MedGemma-4B Haque et al. 2026 — MedConcept (LLM-as-judge)

Llama 3.1 8B

General-purpose instruction-tuned LLM. Defaults to **Uncertain** (~96% on IU X-Ray), showing low discriminative power on small datasets.

MedGemma vs. Llama

3.1: Aligned Rates.

Extracting from the exhaustive evaluation on IU X-Ray (1,508 pairs).
MedGemma rates nearly everything as "Aligned", even random assignments,
failing the null control. Llama 3.1 is highly uncertain overall.

Explanation Method	MedGemma 4B	Llama 3.1 8B
Baseline (512-d)	81.17%	0.86%
SAE Hidden (768-d)	76.26%	0.46%
SPLiCE	81.56%	3.25%
Null (Random)	81.17%	2.65%

Why the judges fail.

Both models lack the discriminative power required for a reliable evaluation on medical domains, but for entirely different reasons.

MedGemma 4B: Catastrophic Positive Bias

It defaults to "Aligned" regardless of semantic content. It approves random noise at ~80% (failing the null control). It is fundamentally incapable of distinguishing true signal from noise in this specific setup.

Llama 3.1 8B: Extreme Uncertainty Bias

Defaults to "Uncertain" (~96% on IU X-Ray). This suggests it either cannot reliably parse the structured verdict format, or it genuinely finds the generated pseudo-reports too ambiguous. Scale helps (better on ROCO v2), but it remains highly conservative.

High computational cost and strong rejection bias.

Running the **26B parameter model** locally via LM Studio proved computationally prohibitive, limiting its evaluation to a single run on the IU X-Ray baseline. However, its results provide crucial insights into cross-model disagreement.

INTER-MODEL DISAGREEMENT (~80PP GAP) ON IU X-RAY BASELINE

JUDGE MODEL	ALIGNED	UNALIGNED	UNCERTAIN	PRIMARY BIAS
Gemma 4 (26B)	1.1%	87.1%	11.9%	Strong Rejection
MedGemma (4B)	81.2%	3.6%	15.2%	Positive Bias
Llama 3.1 (8B)	0.9%	3.1%	96.0%	Uncertainty Bias

Judge behavior across data scales.

How does the 10× scale of ROCov2 affect the LLM-Judge? MedGemma remains positively biased, but Llama 3.1 becomes significantly more decisive as explanation quality improves.

JUDGE PERFORMANCE: IU X-RAY VS. ROCOV2

DATASET	N_PAIRS	MEDGEMMA ALIGNED	LLAMA 3.1 ALIGNED
IU X-Ray	1,508	81.17%	0.86%
ROCOv2	15,957	88.33%	23.14%

MedGemma bias persists

Remains inflated (~81% → 88%), confirming its lack of discriminative power.

Llama 3.1 is sensitive to scale

Jumps from ~1% to 23%. Better explanations (fewer dead features, better naming on ROCov2) unlock more decisive verdicts.

Dense, but weakly separated.

We cluster the active vocabulary into **concept families** with ontology ancestors (RadLex for IU X-Ray, MeSH for ROCOV2) and two degeneracy guards — leaf-root rejection, and a subtree canopy that rejects overly generic ancestors (e.g. "anatomical entity"). SPLiCE, the densest, cuts concepts-per-image **~1.75×** — but silhouette is weak (~ 0.02): the noisy vocabulary makes every cluster inherit a noisy ancestor.

Impl: `src/concept_discovery/organize.py` – `cluster_concepts()`, `annotate_radlex()`

A small upper tail is genuinely faithful.

Do **any** features track real pathology? Correlating activations against the ground-truth labels (with a shuffle null), a **small upper tail** is genuinely faithful — the strongest looks like a *mass*, clinically plausible. But those same features stay **unstable across seeds**: partially faithful, entirely non-reproducible.

Impl: `scripts/run_faithfulness.py`

81.6%

SPLiCE MedGemma · 79.6% null-k

SPLiCE scores **high local plausibility** under a medical-tuned judge — but **almost nothing** under a general one. **Random-k null explanations score comparably** (79.6%), indicating weak discriminability. The Aligned rate is a property of the **judging model** as much as of the explanation. And this local plausibility is **orthogonal** to global identifiability.

Three lessons that generalise.

Null + permutation-invariant

Interpretability claims about discovered concepts must be calibrated against analytical nulls **and** a permutation-invariant statistic. Slot-wise overlap is degenerate.

Partially-negative evaluation

An honest, partially-negative evaluation is more useful than a curated one. The defect is localised (representation site), not global.

4 contributions: relabeling control · matched metric + null · scale refutation · partially-negative eval

Scale is the last suspect

For a factorisation with limited identifiability, data scale is the least likely culprit: 10× changed training health, not identifiability.

Future work

- (i) **The LLM-Judge is model dependant.** The %Aligned rate is a property of the judging model as much as of the concepts. Future work: multi-judge reporting or a judge-agnostic protocol.
- (ii) **Decisive test of our diagnosis: an SAE on the patch-token residual stream.** Decomposing BiomedCLIP's per-patch residual stream at a mid layer (as the SAE literature does) is the experiment that tests whether the 2D limit is intrinsic or escapable.
- (iii) **PadChest.** ROCov2 is 10× larger but also broader (multimodal); PadChest (~160k images, same chest domain as IU X-Ray) is the test of whether scale alone restores reproducibility — begun, not finished.

Thank you · questions — Marc'Antonio Lopez · Nicolò Colle · Carmine Francesco Benvenuto